

Report: Ethical Hacking Lab

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Activity 1 : Wi-Fi -hacking

- use the link: https://www.aircrack-ng.org/doku.php?id=wpa_capture , there download pacp file from there
- Open that file in the wireshark

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Netgear_7e:40:80	Broadcast	802.11	109	Beacon frame, SN=266, FN=0, Flags=....., BI=100, SSID="teddy"
2	8.245211	Netgear_88:ac:82	Broadcast	802.11	42	Probe Request, SN=0, FN=0, Flags=....., SSID=Wildcard (Broadcast)
3	8.246269	Netgear_7e:40:80	Netgear_88:ac:82	802.11	103	Probe Response, SN=355, FN=0, Flags=....., BI=100, SSID="teddy"
4	8.499740	Netgear_88:ac:82	Netgear_7e:40:80	802.11	30	Authentication, SN=0, FN=0, Flags=.....
5	8.502334	Netgear_7e:40:80	Netgear_88:ac:82	802.11	30	Authentication, SN=359, FN=0, Flags=.....
6	8.503836	Netgear_88:ac:82	Netgear_7e:40:80	802.11	79	Association Request, SN=1, FN=0, Flags=....., SSID="teddy"
7	8.505918	Netgear_7e:40:80	Netgear_88:ac:82	802.11	60	Association Response, SN=360, FN=0, Flags=.....
8	8.510014	Netgear_7e:40:80	Netgear_88:ac:82	EAPOL	131	Key (Message 1 of 4)
9	8.514115	Netgear_88:ac:82	Netgear_7e:40:80	EAPOL	155	Key (Message 2 of 4)
10	8.517695	Netgear_7e:40:80	Netgear_88:ac:82	EAPOL	155	Key (Message 3 of 4)
11	8.517699	Netgear_88:ac:82	Netgear_7e:40:80	EAPOL	131	Key (Message 4 of 4)
12	8.521278	Netgear_7e:40:80	Netgear_88:ac:82	802.11	183	Data, SN=363, FN=0, Flags=p....F.
13	8.521795	Netgear_88:ac:82	Netgear_7e:40:80	802.11	151	Data, SN=4, FN=0, Flags=p.....T
14	9.516157	Netgear_7e:40:80	Netgear_88:ac:82	802.11	183	Data, SN=374, FN=0, Flags=p....F.
15	9.516163	Netgear_88:ac:82	Netgear_7e:40:80	802.11	151	Data, SN=5, FN=0, Flags=p.....T

- We get a list of traffic captured
- Use the filter : wlan to know if there is any traffic under wifi

wlan						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Netgear_7e:40:80	Broadcast	802.11	109	Beacon frame, SN=266, FN=0, Flags=....., BI=100, SSID="teddy"
2	8.245211	Netgear_88:ac:82	Broadcast	802.11	42	Probe Request, SN=0, FN=0, Flags=....., SSID=Wildcard (Broadcast)
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5	8.502334	Netgear_7e:40:80	Netgear_88:ac:82	802.11	30	Authentication, SN=359, FN=0, Flags=.....
6	8.503836	Netgear_88:ac:82	Netgear_7e:40:80	802.11	79	Association Request, SN=1, FN=0, Flags=....., SSID="teddy"
7	8.505918	Netgear_7e:40:80	Netgear_88:ac:82	802.11	60	Association Response, SN=360, FN=0, Flags=.....
8	8.510014	Netgear_7e:40:80	Netgear_88:ac:82	EAPOL	131	Key (Message 1 of 4)
9	8.514115	Netgear_88:ac:82	Netgear_7e:40:80	EAPOL	155	Key (Message 2 of 4)
10	8.517695	Netgear_7e:40:80	Netgear_88:ac:82	EAPOL	155	Key (Message 3 of 4)
11	8.517699	Netgear_88:ac:82	Netgear_7e:40:80	EAPOL	131	Key (Message 4 of 4)
12	8.521278	Netgear_7e:40:80	Netgear_88:ac:82	802.11	183	Data, SN=363, FN=0, Flags=p....F.
13	8.521795	Netgear_88:ac:82	Netgear_7e:40:80	802.11	151	Data, SN=4, FN=0, Flags=p.....T
14	9.516157	Netgear_7e:40:80	Netgear_88:ac:82	802.11	183	Data, SN=374, FN=0, Flags=p....F.
15	9.516163	Netgear_88:ac:82	Netgear_7e:40:80	802.11	151	Data, SN=5, FN=0, Flags=p.....T

- Here we can see the SSID "teddy" which means it is the network SSID
- There we see the 2 MAC ADDRESS : Netgear_7e:40:80 & Netgear_88:ac:82 , this is the MAC of the communicating devices.
- Various WiFi frames such as Beacon, Probe, Authentication, Association, and Data frames were observed.
- It uses WPA-4 handshake, to check this use the filter : eapol

eapol						
No.	Time	Source	Destination	Protocol	Length	Info
8	8.510014	Netgear_7e:40:80	Netgear_88:ac:82	EAPOL	131	Key (Message 1 of 4)
9	8.514115	Netgear_88:ac:82	Netgear_7e:40:80	EAPOL	155	Key (Message 2 of 4)
10	8.517695	Netgear_7e:40:80	Netgear_88:ac:82	EAPOL	155	Key (Message 3 of 4)
11	8.517699	Netgear_88:ac:82	Netgear_7e:40:80	EAPOL	131	Key (Message 4 of 4)

- A dictionary attack was performed using Aircrack-ng with the rockyou.txt wordlist
- `aircrack-ng -w /usr/share/wordlists/rockyou.txt wpa.full.cap`

```
(kali@kali)-[~/Downloads]
$ aircrack-ng -w /usr/share/wordlists/rockyou.txt wpa.full.cap
Reading packets, please wait...
Opening wpa.full.cap
Read 15 packets.

# BSSID      ESSID      Encryption
1 00:14:6C:7E:40:80 teddy      WPA (1 handshake)

Choosing first network as target.

Reading packets, please wait...
Opening wpa.full.cap
Read 15 packets.

1 potential targets

Aircrack-ng 1.7

[00:00:10] 69561/14344392 keys tested (6911.75 k/s)

Time left: 34 minutes, 25 seconds      0.48%

KEY FOUND! [ 44445555 ]

Master Key      : 17 4F E9 A8 9F 52 85 FF 0B 7F A3 05 03 DB 38 93
                  75 15 D2 0B CE 17 D8 E2 EE 36 90 F0 47 B4 C5 0E

Transient Key   : 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
                  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
                  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
                  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

EAPOL HMAC     : AE 83 8A AD 75 5C 16 1D 08 87 CD 2C F3 8C AE 60
```

- There we found the KEY : 44445555
- This is the password

Activity 2 : Bluetooth hacking

- Download the pcap file, open it in the wireshark

No.	Time	Source	Destination	Protocol	Length	Info
58	146.838475	controller	host	HCI_EVT	7	Rcvd Command Status (Change Connection Packet Type)
59	146.838486	host	controller	HCI_CMD	14	Sent Remote Name Request
60	146.839475	controller	host	HCI_EVT	8	Rcvd Connection Packet Type Changed
61	146.853473	controller	host	HCI_EVT	7	Rcvd Command Status (Remote Name Request)
62	146.853490	host	controller	HCI_CMD	14	Sent Remote Name Request
63	146.866471	controller	host	HCI_EVT	7	Rcvd Command Status (Remote Name Request)
64	146.887468	MurataManufa_07:2e:...	localhost ()	L2CAP	17	Rcvd Connection Request (SDP, SCID: 0x0040)
65	146.887492	localhost ()	MurataManufa_07:2e:...	L2CAP	21	Sent Connection Response - Success (SCID: 0x0040, DCID: 0x0040)
66	146.902464	controller	host	HCI_EVT	8	Rcvd Number of Completed Packets
67	146.907464	MurataManufa_07:2e:...	localhost ()	L2CAP	17	Rcvd Configure Request (DCID: 0x0040)
68	146.907478	localhost ()	MurataManufa_07:2e:...	L2CAP	19	Sent Configure Response - Success (SCID: 0x0040)
69	146.907483	localhost ()	MurataManufa_07:2e:...	L2CAP	17	Sent Configure Request (DCID: 0x0040)
70	146.932458	controller	host	HCI_EVT	258	Rcvd Remote Name Request Complete
71	146.933470	controller	host	HCI_EVT	7	Rcvd Command Status (No Operation)
72	146.933492	host	controller	HCI_CMD	14	Sent Remote Name Request
73	146.943459	controller	host	HCI_EVT	8	Rcvd Number of Completed Packets
74	146.948458	controller	host	HCI_EVT	8	Rcvd Number of Completed Packets
75	146.953459	MurataManufa_07:2e:...	localhost ()	L2CAP	19	Rcvd Configure Response - Success (SCID: 0x0040)
76	146.954470	MurataManufa_07:2e:...	localhost ()	HCI_ACL	22	Rcvd [Reassembled in #77]
77	146.955458	MurataManufa_07:2e:...	localhost ()	SDP	13	Rcvd Service Search Attribute Request : Headset: Attribute Range (0x0000 - 0
78	146.955738	localhost ()	MurataManufa_07:2e:...	SDP	19	Sent Service Search Attribute Response
79	146.976008	controller	host	HCI_EVT	258	Rcvd Remote Name Request Complete
80	146.977461	controller	host	HCI_EVT	7	Rcvd Command Status (Remote Name Request)
81	146.977485	host	controller	HCI_CMD	14	Sent Remote Name Request
82	146.996450	controller	host	HCI_EVT	8	Rcvd Number of Completed Packets
83	147.003451	MurataManufa_07:2e:...	localhost ()	L2CAP	17	Rcvd Disconnection Request (SCID: 0x0040, DCID: 0x0040, PSM: 0x0001, Service

- Use the filter : bthci_evt , we will get to see the Bluetooth discovery + connection process

No.	Time	Source	Destination	Protocol	Length	Info
35	97.714677	controller	host	HCI_EVT	7	Rcvd Disconnect Complete
36	97.716671	controller	host	HCI_EVT	7	Rcvd Command Status (No Operation)
38	97.726672	controller	host	HCI_EVT	7	Rcvd Command Status (Remote Name Request)
39	118.204249	controller	host	HCI_EVT	258	Rcvd Remote Name Request Complete
40	118.205271	controller	host	HCI_EVT	7	Rcvd Command Status (No Operation)
42	118.255252	controller	host	HCI_EVT	7	Rcvd Command Status (Remote Name Request)
44	136.197254	controller	host	HCI_EVT	7	Rcvd Command Status (Inquiry)
45	138.709830	controller	host	HCI_EVT	258	Rcvd Remote Name Request Complete
46	146.471540	controller	host	HCI_EVT	13	Rcvd Connect Request
48	146.484535	controller	host	HCI_EVT	7	Rcvd Command Status (Accept Connection Request)
49	146.539525	controller	host	HCI_EVT	9	Rcvd Link Key Request
51	146.569520	controller	host	HCI_EVT	13	Rcvd Command Complete (Link Key Request Reply)
52	146.797483	controller	host	HCI_EVT	14	Rcvd Connect Complete
54	146.798483	controller	host	HCI_EVT	10	Rcvd Page Scan Repetition Mode Change
55	146.821478	controller	host	HCI_EVT	9	Rcvd Command Complete (Write Link Policy Settings)
57	146.837474	controller	host	HCI_EVT	6	Rcvd Max Slots Change

- Use the filter: bthci_acl, we will get to see the

No.	bthci_acl	Source	Destination	Protocol	Length	Info
64	146.887468	MurataManufa_07:2e:...	localhost ()	L2CAP	17	Rcvd Connection Request (SDP, SCID: 0x0040)
67	146.907464	MurataManufa_07:2e:...	localhost ()	L2CAP	17	Rcvd Configure Request (DCID: 0x0040)
75	146.953459	MurataManufa_07:2e:...	localhost ()	L2CAP	19	Rcvd Configure Response - Success (SCID: 0x0040)
76	146.954470	MurataManufa_07:2e:...	localhost ()	HCI_ACL	22	Rcvd [Reassembled in #77]
77	146.955458	MurataManufa_07:2e:...	localhost ()	SDP	13	Rcvd Service Search Attribute Request : Headset: Attribute Range (0x0000 - 0xffff)
83	147.003451	MurataManufa_07:2e:...	localhost ()	L2CAP	17	Rcvd Disconnection Request (SCID: 0x0040, DCID: 0x0040, PSM: 0x0001, Service: SDP)
65	146.887492	localhost ()	MurataManufa_07:2e:...	L2CAP	21	Sent Connection Response - Success (SCID: 0x0040, DCID: 0x0040)
68	146.907478	localhost ()	MurataManufa_07:2e:...	L2CAP	19	Sent Configure Response - Success (SCID: 0x0040)
69	146.907483	localhost ()	MurataManufa_07:2e:...	L2CAP	17	Sent Configure Request (DCID: 0x0040)
78	146.955738	localhost ()	MurataManufa_07:2e:...	SDP	19	Sent Service Search Attribute Response
84	147.003474	localhost ()	MurataManufa_07:2e:...	L2CAP	17	Sent Disconnection Response (SCID: 0x0040, DCID: 0x0040, PSM: 0x0001, Service: SDP)

- In the filter : bthci_evt, we can see about inquiry, remote name request, connection request, pin request and connection complete.
- Under the filter : bthci_acl , we will see the L2CAP as protocol and info RcVD connection request.
- When we expand this we will see the source BD_ADDR under the Bluetooth HCI ACL packet.

```

▼ Bluetooth
  [Source: MurataManufa_07:2e:fa (00:0e:6d:07:2e:fa)]
  [Destination: 00:00:00_00:00:00 (00:00:00:00:00:00)]
▼ Bluetooth HCI H4
  [Direction: Rcvd (0x01)]
  HCI Packet Type: ACL Data (0x02)
▼ Bluetooth HCI ACL Packet
  .... 0000 0010 1010 = Connection Handle: 0x02a
  ..10 .... .... .... = PB Flag: First Automatically Flushable Packet (2)
  00.. .... .... .... = BC Flag: Point-To-Point (0)
  Data Total Length: 12
  Data
    [Connect in frame: 52]
    [Disconnect in frame: 89]
    [Source BD_ADDR: MurataManufa_07:2e:fa (00:0e:6d:07:2e:fa)]
    [Source Device Name: ]
    [Source Role: Central (1)]
    [Destination BD_ADDR: 00:00:00_00:00:00 (00:00:00:00:00:00)]
    [Destination Device Name: ]
    [Destination Role: Peripheral (2)]
    [Last Role Change in Frame: 47]
    [Current Mode: Active Mode (0)]
    [Last Mode Change in Frame: 52]
  ▶ Bluetooth L2CAP Protocol

```

- This shows that actual Bluetooth connection ID at HCI level
- Source BD_ADDR: MurataManufa_07:2e:fa --- This is the device address
- Source Role: Central , Destination Role: Peripheral – The communication occurs between a Central device and a Peripheral device.